

INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

M.Tech in Integrated Circuit Design and Technology (ICDT)

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Subject Name:	Analog IC Design Lab
Subject Code:	EE 653
Assignment No:	05
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Instructor Name:	Professor Madhav Pathak

TASK 1: Designing a Fully Differential Amplifier with CMFB

Design a fully differential single-stage amplifier with DC Gain > 45 dB, Power < 0.1 mW, CMRR > 80 dB, Voltage Swing > 1.2 V, UGB > 1 MHz, capacitive loads of 10 pF.

1. Use resistors to set the output common-mode feedback voltage. Justify the value of the used resistance. State the obtained output common-mode voltage (refer to Fig. 1).
2. Design common-mode feedback using source followers to eliminate the loading effect of resistors (refer to Fig. 2)
3. Modify the common mode feedback circuit in (b) by adding an OTA in negative feedback and setting the common mode output voltage to $V_{DD}/2$ (refer to Fig. 3).

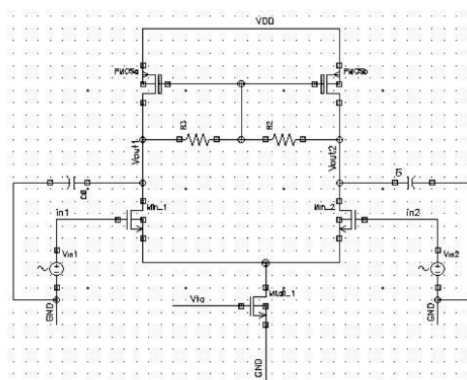


Fig.1

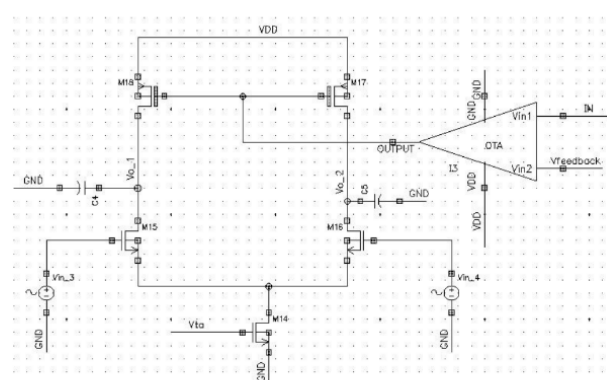


Fig. 3

Figure 1

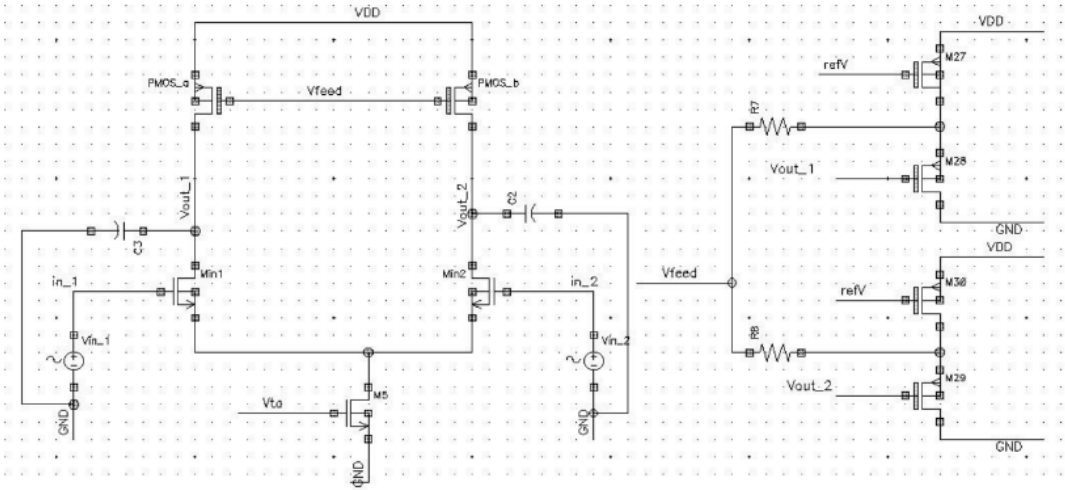


Fig. 2

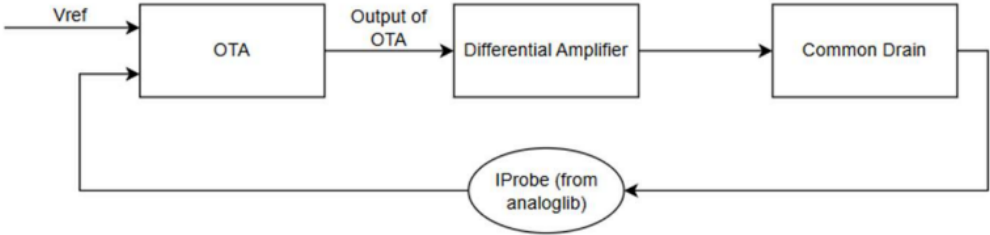


Fig. 4

Figure 2

Figure 3: Schematic with DC parameters

Table 1: Transistor Dimensions

Transistor Name	Type	Width (w)	Length (l)
M7(Load)	PMOS	20.0 μm	1.0 μm
M8(Load)	PMOS	20.0 μm	1.0 μm
M5(Input)	NMOS	6.5 μm	1.0 μm
M6(Input)	NMOS	6.5 μm	1.0 μm
M4(Tail)	NMOS	8.33 μm	1.0 μm

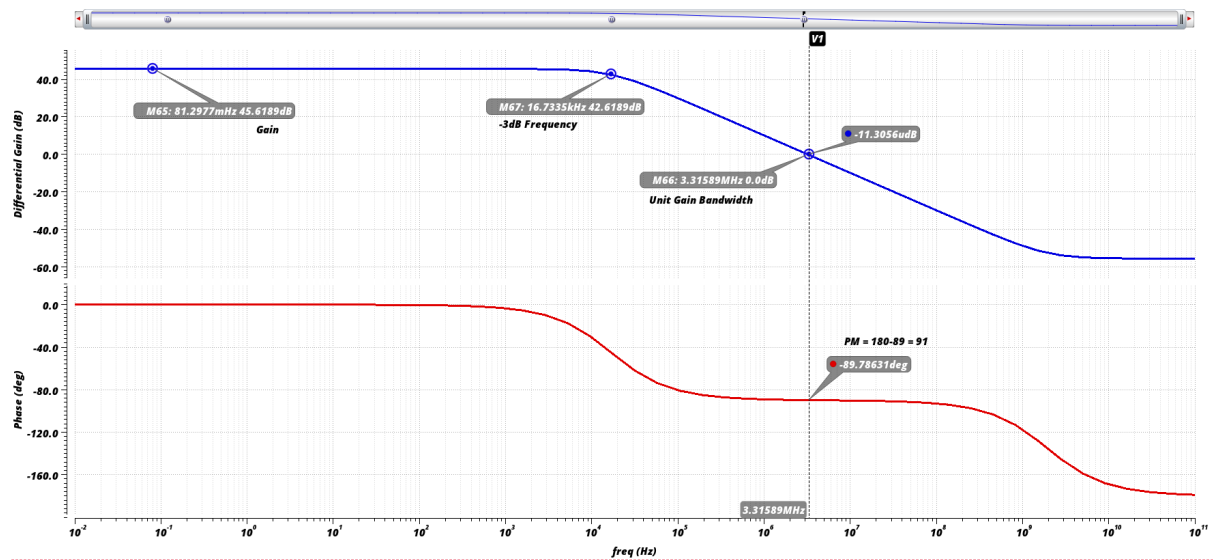


Figure 4: Differential Gain

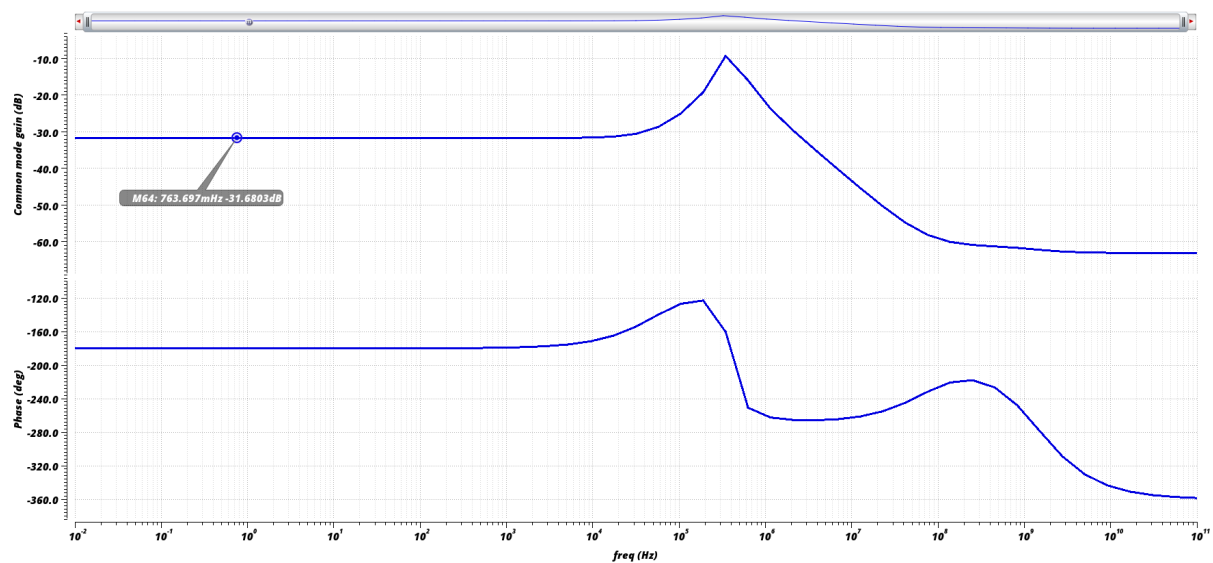


Figure 5: Common Mode Gain

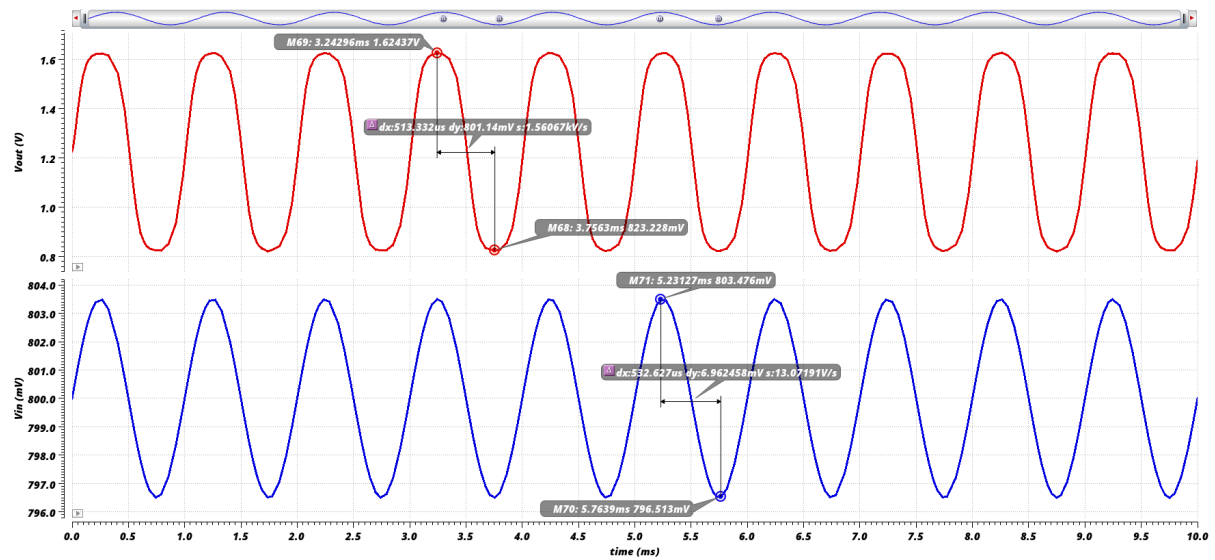


Figure 6: Transient (Swing)

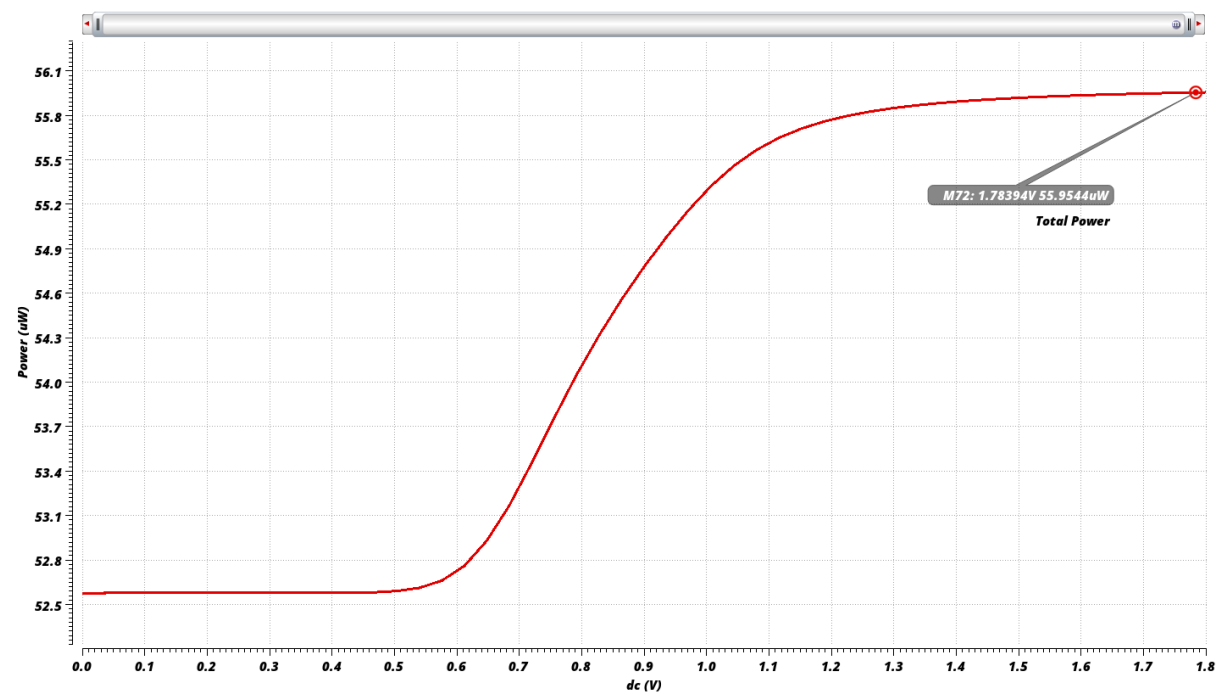


Figure 7: Total Power

Table 2: Transistor DC Parameters

Transistor	I_{ds}	V_{gt}	V_{ds}	g_{ds}	Region
M4 (Tail)	$30.064\mu\text{A}$	152.836mV	218.315mV	$10.308\mu\text{S}$	2
M5 (Input)	$15.033\mu\text{A}$	108.252mV	1.0079V	288.42nS	2
M6 (Input)	$15.033\mu\text{A}$	108.252mV	1.0079V	288.42nS	2
M7 (Load)	$-15.032\mu\text{A}$	-160.375mV	-573.784mV	720.089nS	2
M8 (Load)	$-15.032\mu\text{A}$	-160.375mV	-573.784mV	720.089nS	2

Conclusion

The design and simulation of the fully differential single-stage amplifier were successfully completed using the SCL 180nm PDK. The following observations were made:

- **Gain Performance:** The amplifier achieved a DC differential gain of **45.62 dB**, which satisfies the requirement of > 45 dB.
- **Common Mode Rejection:** The common-mode gain was measured at **-31.68 dB**. This results in a CMRR of approximately **77.3 dB**, showing strong rejection of common-mode signals.
- **Power Efficiency:** The total power consumption was recorded as **55.95 μW** . This is significantly lower than the maximum allowed limit of 0.1 mW, making the design power-efficient.
- **Output Swing:** Transient analysis shows an output peak-to-peak swing of **801.14 mV**.
- **Frequency Response:** The Unity Gain Bandwidth (UGB) was found to be **3.31 MHz**. This exceeds the 1 MHz specification.
- **Stability:** The system is highly stable with a Phase Margin of **91°**.

Task-1: b

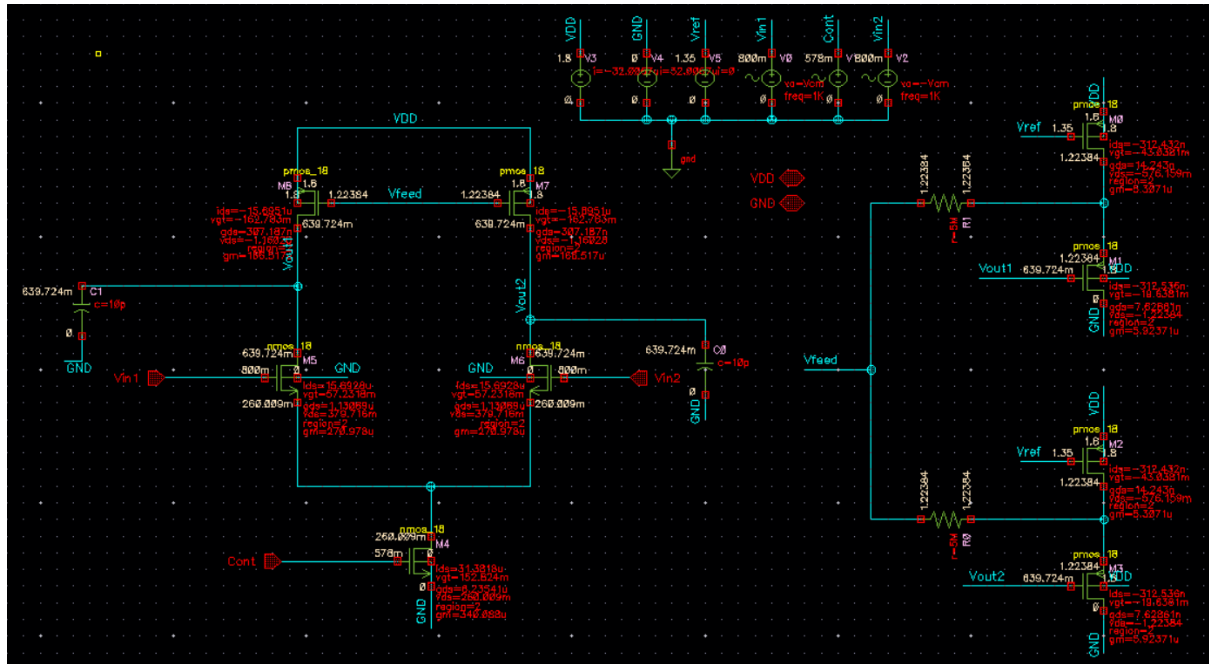


Figure 8: Schematic with DC parameters

Table 3: Transistor Widths and Lengths

Transistor	Type	Width (W)	Length (L)
M4 (Tail)	NMOS	$8.33\mu\text{m}$	$1.0\mu\text{m}$
(M5, M6) (Input)	NMOS	$6.5\mu\text{m}$	$1.0\mu\text{m}$
(M7, M8) (Load)	PMOS	$20.0\mu\text{m}$	$1.0\mu\text{m}$

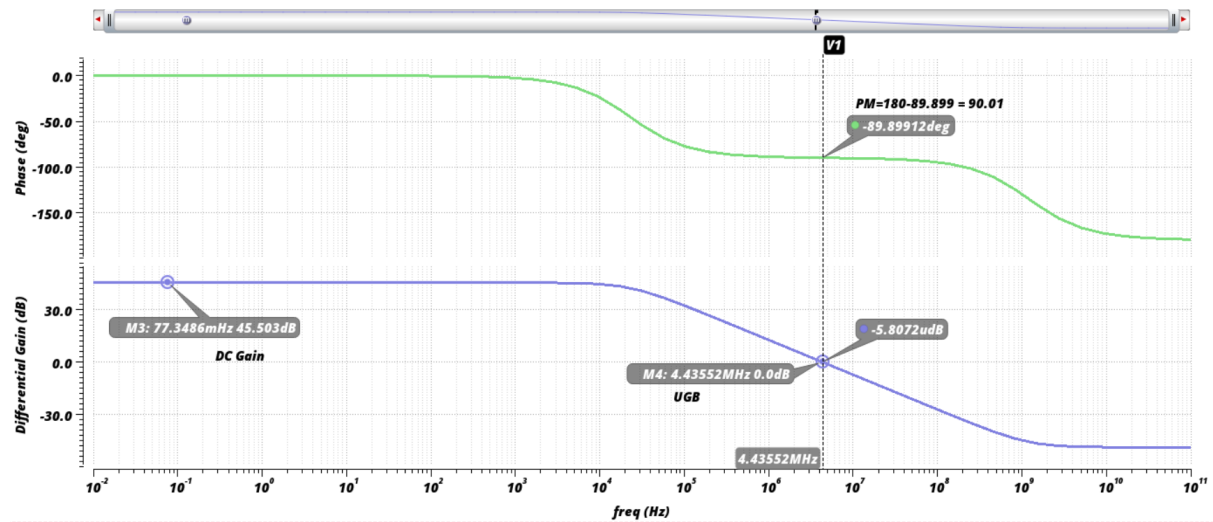


Figure 9: Differential Gain

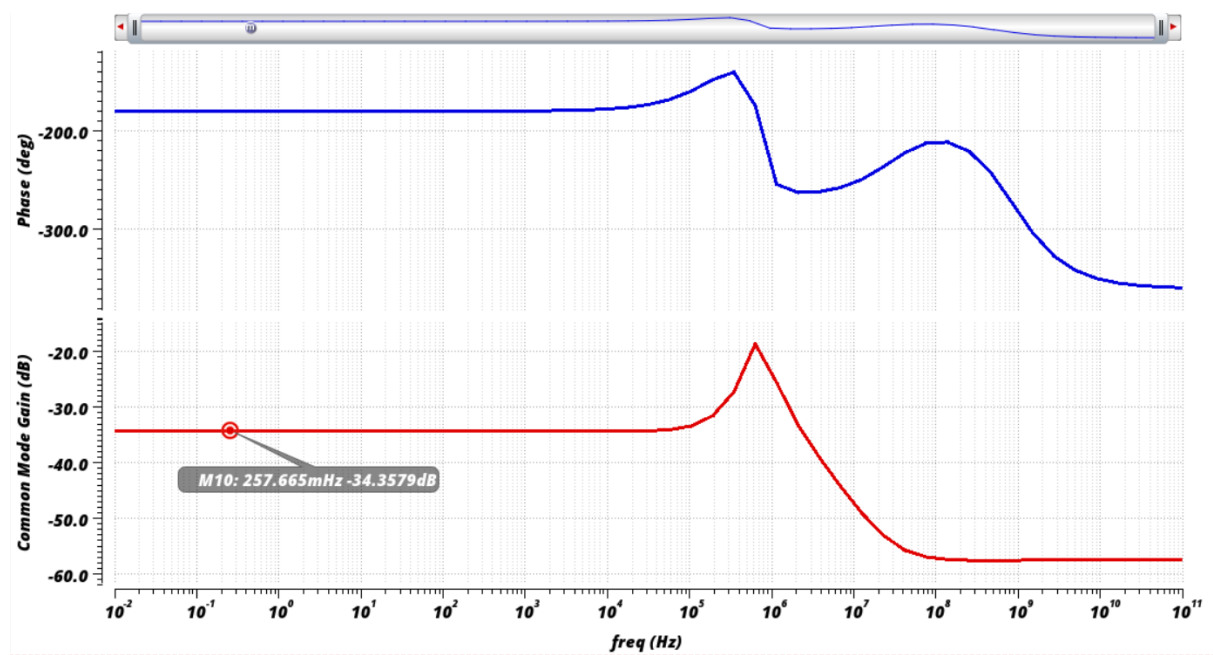


Figure 10: Common Mode Gain

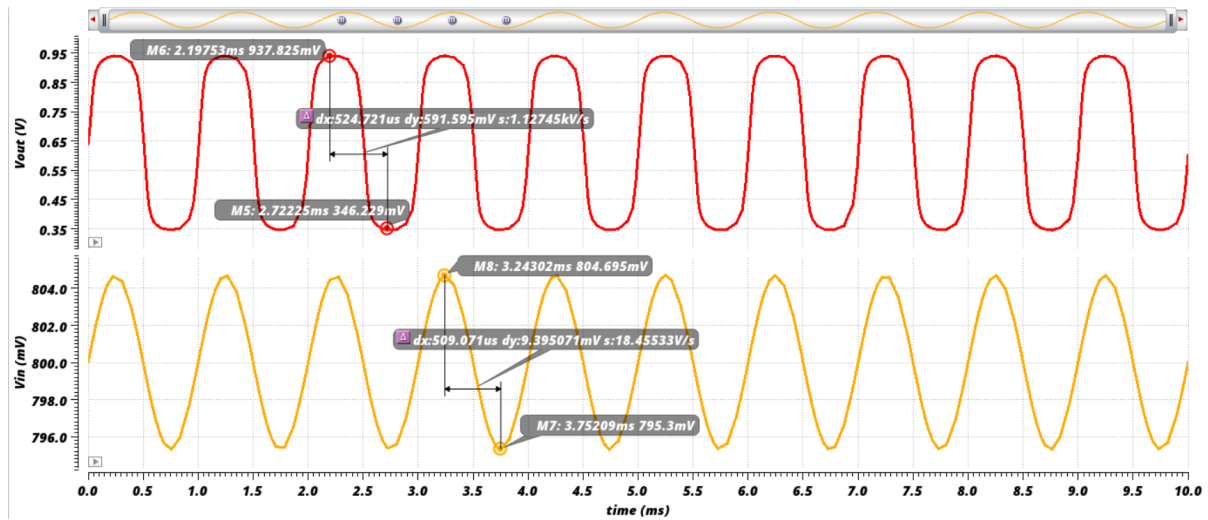


Figure 11: Transient (Swing)

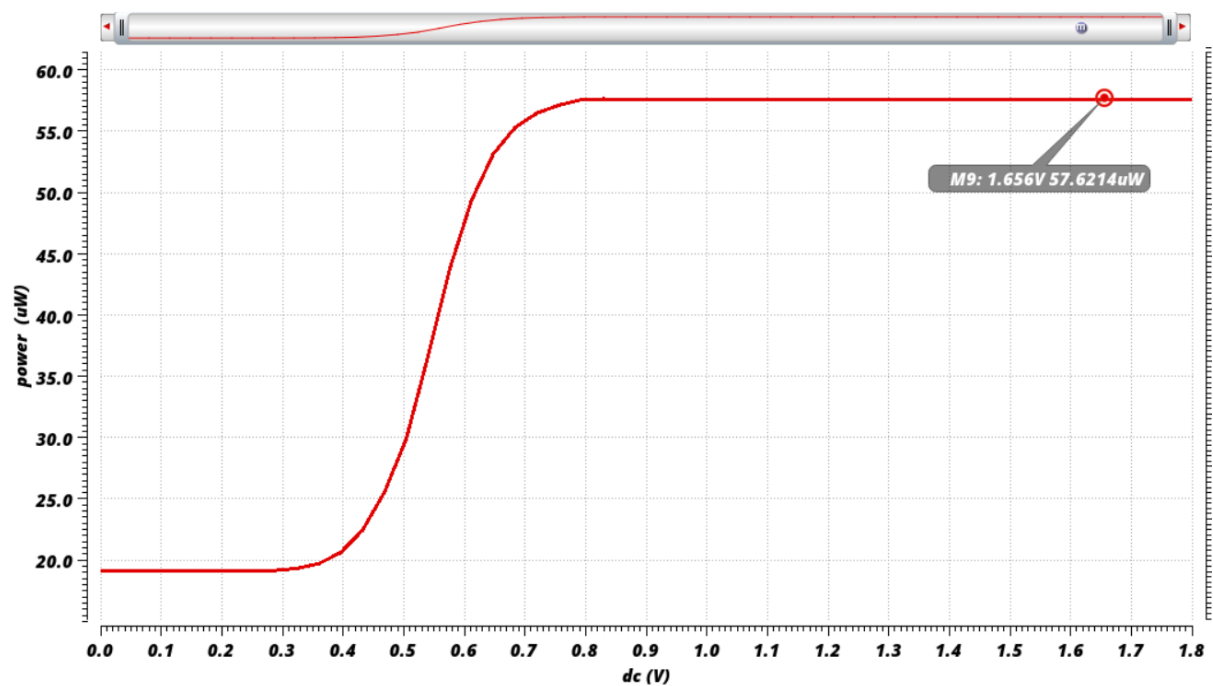


Figure 12: Total Power

Table 4: Transistor DC Operating Parameters

Transistor Name	I_{ds}	V_{gt}	V_{ds}	g_{ds}	Region
M4 (Tail)	$31.381\mu\text{A}$	152.624 mV	260.009 mV	$8.235\mu\text{S}$	2
M5, M6 (Input Pair)	$15.692\mu\text{A}$	57.231 mV	379.716 mV	$1.130\mu\text{S}$	2
M7, M8 (Load Pair)	$-15.695\mu\text{A}$	-162.783 mV	-1.160 V	307.187 nS	2
M0, M2 (SF Bias)	$-312.432\mu\text{A}$	-43.032 mV	-576.159 mV	$14.243\mu\text{S}$	2
M1, M3 (SF Input)	$-312.536\mu\text{A}$	-19.638 mV	-1.223 V	$7.626\mu\text{S}$	2

Conclusion

The design and simulation of the fully differential single-stage amplifier with source-follower common-mode feedback (Task 1b) were completed. The following observations were made:

- **Gain Performance:** The amplifier achieved a DC differential gain of **45.50 dB**, successfully meeting the design target.
- **Common Mode Rejection:** With a common-mode gain of **-34.36 dB**, the design demonstrates robust rejection of common-mode signals.
- **Power Efficiency:** The total power consumption was measured at **57.62 μW** , which is well within the specified **0.1 mW** limit.
- **Output Swing:** The transient response shows an output peak-to-peak swing of **591.60 mV**.
- **Frequency Response and Stability:** The Unity Gain Bandwidth (UGB) was found to be **4.44 MHz** with a highly stable Phase Margin of **90.01°**.

Overall, the integration of source followers in the common-mode feedback path effectively maintains high gain and stability while keeping power consumption low.